

CALEB JOSEPH

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EDUCATION

Toronto Metropolitan University

Sep. 2021 – Apr. 2026 (Exp.)

Bachelor of Engineering, Computer Engineering (Co-op)

Toronto, ON

- **Relevant Courses:** Digital Systems, Computer Architecture, Embedded Systems, Microprocessors, DSA, OOP
- **Affiliations:** National Society of Black Engineers, ColorStack, Career Pathways Program, FEAS

TECHNICAL SKILLS

Languages & Scripting: Verilog, SystemVerilog, Python, C, C++, VHDL, Perl, Bash/CSH, TCL

Tools & Technologies: UVM, VCS, CDC, Regression Testing, Lint, Synthesis, Verdi, Intel Quartus II, Coverage, Cron

Other: RTL Design, Scripting & Automation, Linux, UNIX, Perforce, Verification, CODEC, FPGA, Video IP, SoC, GPU, CPU

EXPERIENCE

Design Verification Intern (Video Domain)

May 2024 – Present

Advanced Micro Devices (AMD) - Python, Scripting, UVM, Verilog, RTL, Automation, Verification

Markham, ON

- Contributing to pre-silicon **video-related IP** verification **regression testing**, learning **SystemVerilog** and **UVM**, ensuring functional coverage and issue identification, covering regression targets by 90%.
- Spearheaded **RTL-level synthesis** (DC ELAB) and **Lint** analysis across video **encoder/decoder** modules, improving **RTL** quality and achieving 30% test coverage uplift.
- Devised and deployed a test configuration extraction tool using **Python** and **RegEx**, accelerating launch and debug cycles by 40%, and reducing build failures in regressions to 0-1 times per week.
- Automated video IP regression testing flows using **Python**, **Perl**, and **CSH** scripts on **Linux** machines, improving regression launch success rate from 70% to 90%.
- Integrated an **automated script** for Clock Domain Crossing (**CDC**) analysis to extract and store **RTL** tile-specific flags for reporting, improving reporting efficiency and reducing error tracking by 15%.
- Implemented a **Python** Module-gated clock-gating script in the power analysis flow, enhancing evaluation efficiency by approximately 20%, streamlining coverage analysis for video IP **RTL** blocks to support a team of 70+ engineers.

Information Technology Intern

May 2023 – Aug. 2023

Envionics Analytics - Python, Scripting, Automation, Linux, Digital Systems, CPU/GPU

Toronto, ON

- Directed donation of **40+** laptops, repurposing **\$10,000+** of equipment leveraging a **Python** script for model identification in **BIOS/OS** to ensure data integrity through secure boot showcasing hardware diagnostics skills.
- Automated inventory cost allocation with a **Python** script for over **300** assets, improving efficiency by 40%.

PROJECTS

RTL Design of Round Robin Arbiter | *SystemVerilog, Verilog, RTL Design, EDA Playground*

GitHub 

- Designed a 4-request, 4-grant **round robin arbiter** using **RTL-level Verilog**, with state-machine scheduling logic.
- Engineered a **SystemVerilog** testbench to simulate and validate the design, verifying correct arbitration behavior.

Multi-stage RISC Pipelined Processor | *VHDL, Quartus II, Altera DE2-115, Cyclone-IV EP4CE115F29C7 FPGA*

GitHub 

- Constructed a 32-bit **3-stage pipeline RISC CPU** using **VHDL** on an Altera DE2-115 **FPGA** board with **Intel Quartus II** for **RTL** synthesis and simulation, achieving a target frequency of **>50MHz**.
- Designed and simulated a register set, **program counter**, **ALU**, data path, and control unit, ensuring optimized **RISC** processing and reducing data processing delays by 15%.

Ray Tracing Application | *C, C++, ImGui, Visual Studio, Walnut Framework*

GitHub 

- Developed a ray tracing application in **C++**; enhancing the Render class and leveraging multi-threading techniques, reducing render times by **30%**, assessing gaming **GPU** acceleration for **CUDA** performance processing uplift.

PROFESSIONAL DEVELOPMENT

Program Manager

May 2024 – Present

Intern Steering Committee (ISC), Advanced Micro Devices (AMD) - Project Management, Collaboration

Markham, ON

- Led events aligned to intern interests, fostering a vibrant community of 200+ co-ops at the **AMD Markham** site.
- Featured in **AMD's Global Intern Spotlight** for contributions as an **ISC Program Manager** and **Design Verification Intern**, promoting inclusion and intern engagement.
- Won **AMD Intern Innovation Showcase** for BurnBot, an **AI** task manager automating task extraction and workflow tracking for burnout prevention and **verification** support.

Open to relocation.

Canadian Citizen.